SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING (5 Questions)

		Course Title:	COMPUTER VISION
Course Code:	ITA05	Assessment:	Assessment Tool 1 – Analytical Problem Solving
CO Assessed:	CO2 – Manipulation(BL3)	Total Marks:	50 (5 Questions × 10 Marks)
Weightage:	20% of CIA		
CO2 Statement:	*Apply image enhancement and segmentation techniques for extracting meaningful regions from images..(BL3)*		
Student Name:	T. V. Sai Jagadeesh	Reg. No.:	19247214 6
Section / Batch:	Slot D/2024	Date:	28/7/26

Code of Conduct

I T.V.S. Jagadeesh / 192472146 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: T. Sai

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Grey Level Transformations	Enhancing image brightness and contrast using negative, log, power-law (gamma), and contrast stretching transformations.	1	10
Histogram Processing Techniques	Improving image contrast through histogram analysis, histogram equalization, and histogram matching.	2	10
Spatial Domain Enhancement	Reducing noise and enhancing image quality using smoothing filters and sharpening filters.	3	10
Frequency Domain Enhancement	Enhancing images by applying low-pass and high-pass filters.	4	10
Image Segmentation	Separating objects from the background using point, line, and edge detection, thresholding, region-based segmentation, edge linking, and boundary detection techniques.	5	10
	GRAND TOTAL:		**50 Marks**

Annexure A – ANALYTICAL PROBLEM SOLVING

1.Grey Level Transformation

A grayscale image is enhanced using a negative transformation.

(a) Explain the purpose of grey level transformation in image enhancement

(b) Apply the negative transformation:

$$s = 255 - r$$

to the following pixel values:

$$r = 50,100,150,200$$

Compute the transformed pixel values and interpret the effect on the image.

2.Histogram Equalization

A 4-bit grayscale image has the following histogram:

Intensity	Frequency
0	4
1	6
2	10
3	6

(a) Explain why histogram equalization is used for image enhancement.

(b) Compute the cumulative distribution function (CDF).

(c) Determine the equalized intensity values and interpret the improvement in image contrast.

3.Gaussian Smoothing

A Gaussian filter is applied to reduce image noise.

(a) Explain why Gaussian filtering is preferred over averaging filtering.

(b) Apply the Gaussian kernel

$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

to the image region

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

Compute the new center pixel value and interpret the smoothing effect.

4.Edge Detection Using Sobel Operator

An image is processed using the horizontal Sobel operator.

(a) Explain the concept and purpose of edge detection.

(b) Apply the horizontal Sobel mask

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

to the image

$$\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

Compute the gradient at the center and interpret the result.

5.Contrast Stretching

A grayscale image has intensity values in the range 50–150.

(a) Explain the need for contrast stretching in image enhancement.

(b) Using linear contrast stretching to map the range [50,150] to [0,255], compute the transformed values for

$$r = 50,100,150$$

Show the calculations and interpret the improvement in image contrast.

CO2 - ATI Analytical Problem Sol. 192472146

43/50

Q1. Gray Level Transformation

a) Purpose of grey level transformation
Gray level transformation is used to modify pixel intensity values in an image

* Contrast
* visibility details
* Overall appreances

b) Negative transformation.
$\therefore s = L-1-r$
$s = 255 - r$

r	s
50	$255-50 = 205$
100	$255-100 = 155$
150	$255-150 = 105$
200	$= 55$

Interpretation:
→ Dark areas becomes bright
→ Bright areas becomes darker

Q2. Histogram Equalization

a) Purpose:-
→ Improve image contrast
→ Spread pixel value across full intensity

b) CDF (Cummulative Distribution frequency)

G: T

Step1:-

Intensity(r)	Frequency	probability $P(r) = \frac{freq}{N}$
0	4	$4/16 = 0.25$
1	2	$2/16 = 0.125$
2	2	0.125
3	8	0.5
	$N = 16$	

PDF ?

Step2:-

$CDF = \sum P(r)$

r	CDF
0	0.25
1	$0.25 + 0.125 = 0.375$
2	$0.375 + 0.125 = 0.5$
3	$0.5 + 0.5 = 1.0$

c)

Step3:- Equalization

$L = 16$

$s = (16 - 1) \times CDF$

r	s
0	$15 \times 0.25 = 3.75$
1	$15 \times 0.375 = 6$
2	8
3	15

Q3. Gaussian Smoothing

→ A gaussian filter is applied to reduce image noise.

a)

Average filter	Gaussian filter
→ Average weight	weighted centre
→ Blure Edge	~~No~~ Preserve edge
→ Less natural	More natural smoothing

Apply Gaussian Kernel

Mask:-

$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

image region:-

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

$$= \frac{1}{16}\left[(1\times10)+(2\times20)+(1\times10)+(2\times20)+(4\times40)+(2\times20)\right.$$
$$\left.+(1\times10)+(2\times20)+(1\times10)\right]$$

$$= \cancel{\frac{1}{16}}\left[\cancel{640}\right]$$

$$= 40$$

Q4:- Edge Detection Using Sobel operator

a) Purpose:-

Edge detection identifies boundaries in images where intensity change sharply.

Uses:-

* Object detection
* Feature extration
* Image segmentation

b) Applying horizontal Sobel mask:-

$$G_x = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \quad \text{Orginal image} = \begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

$= G_x \times O.img$

$= [-10 - 20 - 10 + 0 + 0 + 0 + 30 + 60 + 30]$

$= [80]$

$$= \begin{bmatrix} 10 & 10 & 10 \\ 20 & 80 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

Q5. Constrast Stretching:-

A gray scale image has intensity values in the range 50 - 150.

a) Contrast stretching in image enhancement

→ Image has limited intensity range.

→ Looks dull or faded

* It expands intensity values into improve visibility.

b) Linear Contrast stretching

$$s = \frac{(r - r_{min})}{(r_{max} - r_{min})} \times 255$$

$r_{min} = 50$, $r_{max} = 150$

if $r = 50$

$$s = \frac{50 - 50}{150 - 50} \times 255$$

$$= 0$$

$r = 100$

$$s = \frac{100 - 50}{150 - 50} \times 255$$

$$= 127.5$$

$$\approx 128$$

$r = 150$

$$s = \frac{\cancel{150 - 50}}{\cancel{150 - 50}} \times 255$$

$s = 255$